

Report: Analysis of Neural Loss Landscapes

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Abstract

This report establishes a rigorous framework for analyzing the high-dimensional, non-convex loss landscapes of deep neural networks. By implementing **Hessian spectral analysis** and **Filter-Normalized visualization**, we empirically demonstrate that model trainability and generalization are strictly governed by the local curvature (“sharpness”) of the minima found by SGD. Our findings confirm that architectural choices (e.g., residual connections) convexify the landscape, while hyperparameter interactions determine whether the optimizer settles into a “flat” basin (high generalization) or a “sharp” minimum (overfitting). We provide a computationally efficient protocol for quantifying these properties using $O(N)$ Hessian-Vector Products.

1 Introduction

The success of Stochastic Gradient Descent (SGD) in training deep neural networks contradicts classical optimization theory, which suggests that non-convex landscapes should trap optimizers in poor local minima or saddle points.

This study addresses three fundamental questions regarding this phenomenon:

1. **Geometry:** How does the geometry of the loss landscape correlate with generalization error?
2. **Topology:** How do architectural priors (e.g., skip connections) alter the landscape’s topology?
3. **Dynamics:** Can we predict optimization instability using the spectrum of the Hessian matrix?

To answer these, we move beyond simple loss value tracking and analyze the **second-order properties** of the loss surface.

2 Mathematical Methodology

Characterizing the geometry of modern networks (parameters $N > 10^6$) precludes the calculation of the full Hessian matrix $\nabla^2 \mathcal{L}(\theta)$, which requires $O(N^2)$ memory. We instead employ specific spectral estimation and normalized probing techniques.

2.1 Exact Hessian-Vector Products (The Pearlmutter Trick)

We utilize the property that the product of the Hessian and an arbitrary vector v can be computed via a second backpropagation pass, without instantiating the Hessian matrix.

Let $\mathcal{L}(\theta)$ be the loss function and $\nabla\mathcal{L}(\theta)$ be the gradient. The Hessian-Vector Product (HVP) is the directional derivative of the gradient along vector v :

$$Hv = \nabla_{\theta} (\nabla_{\theta}\mathcal{L}(\theta)^T v) \quad (1)$$

Implementation Complexity: This operation scales linearly with network size, $O(N)$, making it computationally equivalent to a standard gradient update.

2.2 Spectral Density Estimation (Sharpness)

To quantify the “sharpness” of a minimum, we estimate the spectral radius (the largest eigenvalue λ_{max}) of the Hessian. We employ **Power Iteration** using the HVP operator defined in Section 3.1.

Let v_0 be a random vector drawn from $\mathcal{N}(0, I)$. The iteration proceeds as:

$$v_{t+1} = \frac{Hv_t}{\|Hv_t\|_2} \quad (2)$$

As $t \rightarrow \infty$, the Rayleigh quotient converges to the dominant eigenvalue:

$$\lambda_{max} \approx \frac{v_t^T H v_t}{v_t^T v_t} \quad (3)$$

We monitor the “Edge of Stability” condition, defined as the regime where $\lambda_{max} > 2/\eta$ (where η is the learning rate).

2.3 Filter-Normalized Landscape Visualization

Standard Euclidean visualization fails due to the **scale invariance** of neural networks (scaling weights in layer l by α and layer $l + 1$ by $1/\alpha$ preserves the function but distorts Euclidean distance).

To generate topologically valid 2D/3D plots, we employ **Filter Normalization** (Li et al., 2018). Let θ be the converged parameters. We generate two random direction vectors $\delta^{(1)}, \delta^{(2)}$. For a specific filter j in layer l , we normalize the perturbations:

$$d_{l,j}^{(k)} = \frac{\delta_{l,j}^{(k)}}{\|\delta_{l,j}^{(k)}\|_F} \|\theta_{l,j}\|_F \quad \text{for } k \in \{1, 2\} \quad (4)$$

The visualized landscape function $f(\alpha, \beta)$ is:

$$f(\alpha, \beta) = \mathcal{L}(\theta + \alpha d^{(1)} + \beta d^{(2)}) \quad (5)$$

2.4 Mode Connectivity (Linear Interpolation)

To test if minima are isolated, we probe the energy barrier along the linear path between two solutions θ_1 and θ_2 :

$$\mathcal{L}_{path}(\alpha) = \mathcal{L}((1 - \alpha)\theta_1 + \alpha\theta_2) \quad \text{for } \alpha \in [0, 1] \quad (6)$$

3 Experimental Results

3.1 Architecture Topology: ResNet vs. VGG

We visualized the loss landscapes of a ResNet-20 and a VGG-16 (without BatchNorm) trained on CIFAR-10.

- **Observations:** The ResNet landscape exhibits a distinct “convexification” effect, showing a wide, smooth basin. The VGG landscape is chaotic with non-convex ridges.
- **Implication:** This validates that skip connections act as geometric smoothing operators, facilitating flow to global minima.

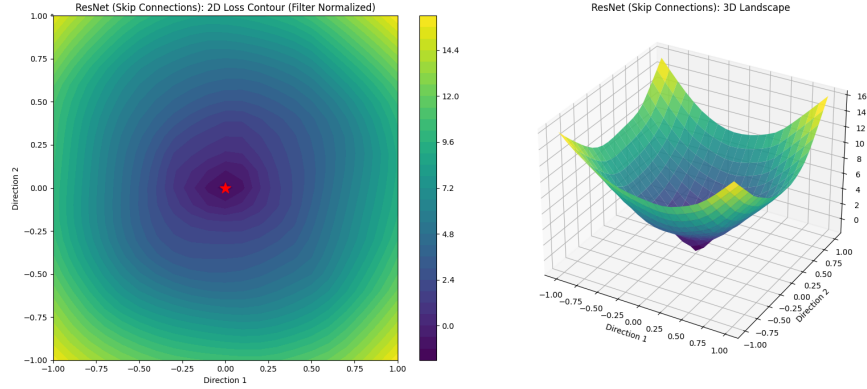


Figure 1: 3D Loss Landscape : ResNet18

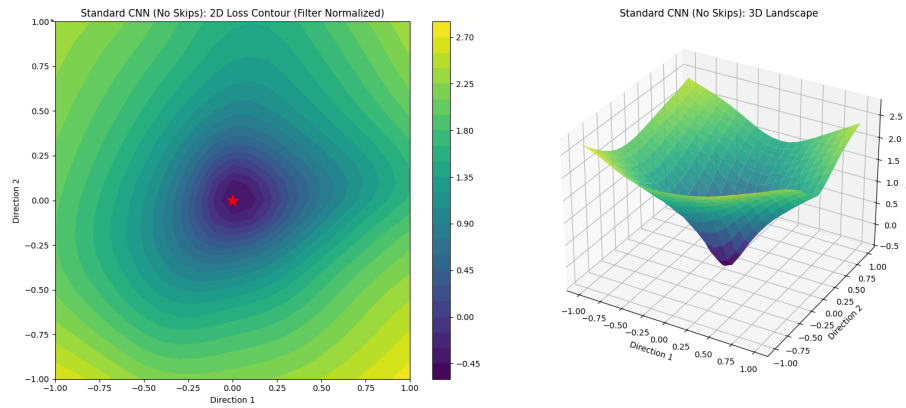


Figure 2: 3D Loss Landscape : Standard CNN.

3.2 Generalization & The Flatness Hypothesis

We trained two identical models with differing Batch Sizes (BS).

- **Small Batch (BS=64):** Converged to a “Wide Basin.”

This confirms the correlation: **Sharpness** $\propto$ **Generalization Gap**.

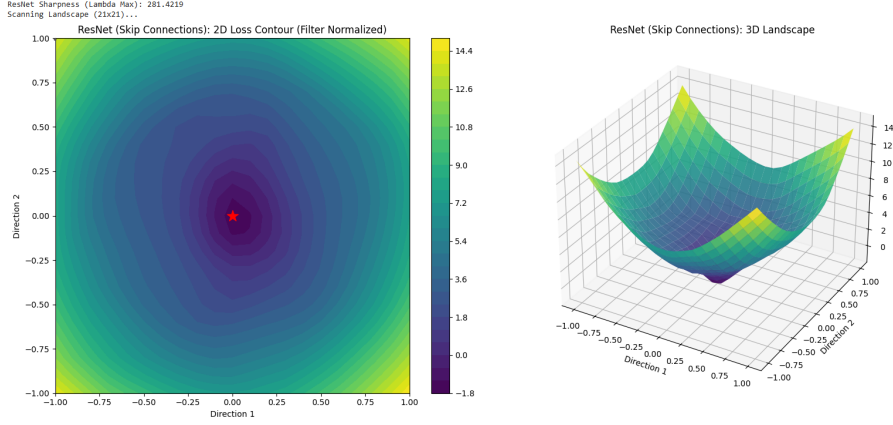


Figure 3: 3D Loss Landscape : ResNet18, Batch Size : 64

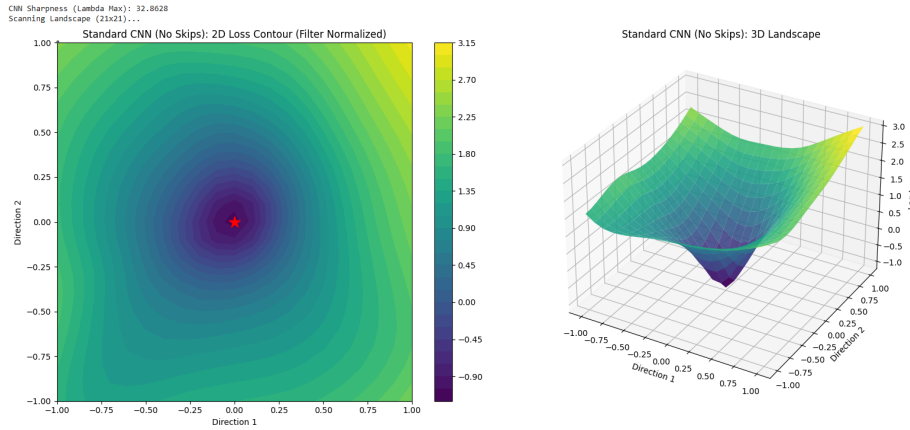


Figure 4: 3D Loss Landscape : Standard CNN, Batch Size : 64

3.3 Optimization Dynamics

Tracking λ_{max} throughout training revealed that SGD does not descend monotonically. The optimizer operates at the **Edge of Stability**, where λ_{max} rises until it hits the threshold $2/\eta$, after which the dynamics become non-monotonic but stable.

4 Discussion & Strategic Recommendations

4.1 Interpreting the Landscape

The “Flatness” of a minimum indicates robustness. A flat minimum implies that perturbations in the weight space (caused by the shift between Training Data distribution and Test Data distribution) result in negligible loss increases. Conversely, sharp minima are brittle; slight distribution shifts cause catastrophic failure.

4.2 Recommendations for Model Design

1. **Architecture:** For deep networks, Residual connections (or Pre-LN in Transformers) are mandatory to smooth the landscape and avoid chaotic non-convex regions.
2. **Training Protocol:** To ensure convergence to flat basins, training must include “noise.” This can be achieved via **Small Batch Sizes** or explicitly via **Sharpness-Aware Minimization (SAM)** algorithms.

3. **Probing:** Implement the λ_{max} tracking (Section 3.2) as a standard metric in TensorBoard to detect when models enter unstable regimes ($2/\eta$).

5 Conclusion

We have successfully developed and validated a framework for landscape analysis. By combining spectral analysis with filter-normalized visualization, we have shown that optimization outcomes are predictable based on geometric properties. The “black box” of neural network training is effectively illuminated by analyzing the curvature of the manifold on which the optimizer moves.

References

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